Certified Correction of Vision–Language Hallucinations: Gains, Preconditions, and a Sequential Failure Mode

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Reviewed on OpenReview: *https://openreview.net/forum?id=XXXXXXXXXX*

Abstract

Selective prediction equips a vision–language model (VLM) with a distribution-free guarantee on the answers it emits, and pays for that guarantee by abstaining on the rest. A recent impossibility result shows the bill cannot be reduced by better engineering: when the base risk μ exceeds the target α, *any* predictor that may only *emit or abstain the model's own answer* must abstain on at least $(\mu - \alpha)/(1 - \alpha)$ of inputs, whatever its score, bound, or calibration budget. This paper takes that premise seriously and asks what changes when the predictor is permitted to emit a *different* answer. We introduce Certified-Correction Risk Control (CCRC): accept the model's answer where a correctness score is high, substitute an answer supplied by an *independent* evidence channel where that channel's evidence is strong, abstain otherwise, and certify the error rate of everything emitted at level α with confidence $1 - \delta$.

We prove validity for CCRC, characterise its behaviour, and evaluate it across five benchmarks, three backbones and a mitigation decoder. Four findings organise the paper. First, the nonconformity *score* matters more than the procedure: a detection-grounded score improves certified coverage over model-internal uncertainty by up to 10.2 points, and we decompose our total margin over a published CLIP-based scorer to show that 62.6 of 67 points come from the score rather than from our algorithm. Second, repairs cannot be self-certified — we prove that certifying a substitution with the complement of the score that flagged it is circular, and confirm that the resulting procedure repairs nothing. Third, where CCRC helps it does so by *dilution*: roughly 1.5% of items repaired at a strict evidence gate buys 3–6× that mass in certified coverage, because high-precision repairs lower the emitted set's average risk and relax the binding constraint; we give a precondition, checkable before calibration, that predicts both the gains and the single dataset on which CCRC loses. Fourth, the natural generative extension fails: in a paired matched-prefix experiment, repairing a claim mid-caption significantly *increases* downstream hallucination ($+0.207 \pm 0.091$ mentions, $n = 121$, $p = 0.025$), which closes off a fixed-point calibration argument and exposes an unpriced cost that any sequential procedure must charge for. We also reproduce the closest concurrent method and find the two mechanisms are complementary rather than competing. Code, extracted per-item signals, and all negative results are released.

1 Introduction

A vision–language model asked what is in a photograph will sometimes answer confidently about things that are not there. The failure is systematic rather than incidental: benchmarks built specifically to elicit it report error rates from a few per cent to well over half, depending on the model and how adversarially the question is posed (Li et al., 2023; Wang et al., 2023; Guan et al., 2024). Consequences scale with the stakes.

A caption generator that invents a handbag is a nuisance; a clinical report generator that invents a finding is a hazard (Xia et al., 2024).

Two research programmes address this, and they are almost disjoint.

The first, much larger, programme *reduces* hallucination. It modifies decoding to penalise over-trust in the language prior (Huang et al., 2024), contrasts distorted and clean visual inputs (Leng et al., 2024; Zhang et al., 2025), intervenes on the attention layers where the failure is localised (Anonymous, 2025a;g;e;d), injects the output of vision experts (Zhao et al., 2025), verifies and rewrites post hoc (Yin et al., 2024; Anonymous, 2025h), or retrieves external knowledge (Lee et al., 2024; Xia et al., 2024). These methods work: reported hallucination rates fall substantially. What they do not provide is a *bound*. A deployer who asks "how often is an emitted claim wrong?" receives an empirical average on a benchmark, not a guarantee that transfers.

The second programme supplies exactly that. Conformal prediction and conformal risk control turn a heuristic score into a decision rule with a finite-sample, distribution-free guarantee (Vovk et al., 2005; Bates et al., 2021; Angelopoulos et al., 2021). Applied to generation, this yields *selective prediction*: emit when a nonconformity score clears a calibrated threshold and abstain otherwise, and the error rate among emitted outputs is at most α with probability $1 - \delta$ (Quach et al., 2024; Mohri & Hashimoto, 2024; Anonymous, 2025b;f). The guarantee is real and the assumptions are mild. The cost is abstention.

The bill for abstention, and a premise worth re-reading. How large is that cost? Anonymous (2026c) answer in closed form, and the answer is discouraging: if the base risk μ exceeds the target α, any such predictor must abstain on at least $(\mu - \alpha)/(1 - \alpha)$ of inputs, *regardless of score quality, bound tightness, or calibration set size*. The floor depends only on quantities known in advance. On a benchmark where a model is wrong a fifth of the time and the target is 10%, at least an eighth of all inputs must be refused before any method is chosen.

The bound is correct, and its premise is stated plainly in the source: the predictor may only *emit or abstain the model's own answer*. It is a premise every conformal method for VLMs we are aware of satisfies. Claim-filtering methods delete unsupported claims (Anonymous, 2025b; Mohri & Hashimoto, 2024). Abstention methods decline to answer (Anonymous, 2025f; 2026e). The closest recent work acquires additional visual evidence and *re-scores* the original assertion, explicitly leaving the answer unchanged (Anonymous, 2026a). In every case the object emitted is the model's own output or nothing.

This paper asks the obvious next question. *What can a selective predictor do if it is allowed to emit a different answer?*

What changes, and what does not. Permitting substitution does not evade the mathematics; it changes the object being certified. If the emitted answer may come from somewhere other than the model, then the quantity to control is the error rate of a *mixture* — accepted model answers plus substituted ones — and the bound of Anonymous (2026c) no longer applies because its premise is violated. The guarantee has to be re-derived for the mixture, which we do (Theorem 1), and the substitution has to come from somewhere trustworthy, which turns out to be a strong constraint rather than an implementation detail (Theorem 2).

It also does not follow that substitution is a good idea. Most of this paper is spent establishing when it is not. Repairs drawn from the wrong source cannot be certified at all. Repairs admitted too liberally destroy coverage. On one of our five benchmarks the whole apparatus is counterproductive, for a reason we can state precisely and check in advance. And in the generative setting — the setting that motivated the idea, since replacing a hallucinated word is more useful than deleting a sentence — correction turns out to make the rest of the caption worse.

We report all of this. The paper is organised so that each design decision is forced by a measurement rather than asserted, and the negative results are presented as findings rather than as caveats appended to a positive headline.

Contributions.

1. **A procedure and its guarantee** (§4). CCRC emits accepted, repaired, or abstained outputs and certifies the risk of the emitted mixture: $\Pr[\mathcal{R}(\mathcal{E}) \leq \alpha] \geq 1 - \delta$ (Theorem 1), using fixed-sequence testing over a nested policy family with exact binomial bounds. We also give the minimum certifiable sample size (Lemma 1), a small result with practical teeth: a fixed sequence started below it aborts and returns zero coverage.

2. **An impossibility result for self-certified repair** (Theorem 2). Certifying a substitution using the complement of the score that flagged the original is circular: for binary decisions the repair's risk equals the model's *accuracy* on the flagged region, so certifiability demands a region of accuracy below α, a strictly stronger requirement than identifying low confidence. Empirically, even the bottom 2% of our score retains 13.3% accuracy against $\alpha = 0.10$, and a version of CCRC built this way repairs essentially nothing. Certified correction therefore *requires* a second, independent evidence channel — a constraint on system architecture, not a modelling preference.

3. **A mechanism and a precondition** (§4.6, §4.7). We show why repair helps when it helps: high-precision repairs lower the emitted set's average risk, relaxing the binding constraint and permitting a more permissive acceptance threshold (Proposition 2). The effect is levered: 0.9–1.8% repaired mass yields 1.9–4.7 points of coverage. We then bound the achievable gain by the abstained mass (Proposition 3), which explains the asymmetry we observe — upside is capped, downside is not — and yields a precondition (μ comfortably above α) that correctly predicts the one benchmark where CCRC loses.

4. **A sequential negative result** (§6). In a paired matched-prefix experiment on free-form captions, repairing a hallucinated claim significantly increases hallucination in the continuation. This rules out the monotone fixed-point argument on which an on-policy sequential guarantee would rest, and we give the weaker coupling bound that survives (Theorem 3).

5. **Honest attribution and positioning** (§5.5). We decompose our margin over a published scorer and report that most of it is the score, not the procedure. We reproduce the closest concurrent method (Anonymous, 2026a) and find the two mechanisms compose rather than compete.

2 Related work

2.1 Hallucination in vision–language models

Phenomenon and taxonomy. Hallucination in multimodal models is conventionally decomposed by what is fabricated: *existence* (an absent object is asserted), *attribute* (a present object is described wrongly), and *relation* (spatial or interactional claims that do not hold) (Wang et al., 2023). The distinction matters for method design, because the evidence needed to refute each differs: existence claims can be checked by a detector, attribute claims need finer perception, and relation claims need structure.

Benchmarks. POPE (Li et al., 2023) probes existence with balanced yes/no questions and three negative samplings — random, popular, and adversarial (co-occurring) — the last being substantially harder because plausible distractors defeat language priors. CHAIR (Rohrbach et al., 2018) scores free-form captions by the fraction of mentioned objects absent from ground truth. AMBER (Wang et al., 2023) provides an LLM-free benchmark spanning generative and discriminative splits with per-image lists of present and hallucinated objects. MMHalBench (Sun et al., 2024b) covers diverse hallucination types in question answering; MME (Fu et al., 2023) measures broad perception and cognition; HallusionBench (Guan et al., 2024) targets visual illusions and reasoning traps and is hard enough to push strong models to near chance. We use POPE, AMBER, MME, GQA (Hudson & Manning, 2019) and HallusionBench, and we note in §5 that the benchmark's *composition* determines whether detector-based evidence is applicable at all.

Mitigation by decoding. A prominent family modifies the decoding distribution. OPERA (Huang et al., 2024) penalises the attention pattern associated with over-trusting the language prior and reallocates attention retrospectively. Visual contrastive decoding (Leng et al., 2024) contrasts logits from clean and diffusion-corrupted images to suppress language-prior-driven tokens; instruction contrastive decoding and related variants use other perturbations. DeGF (Zhang et al., 2025) synthesises an image from the candidate response and diffs it against the input. Attention-level analyses locate the failure in specific layers and heads and intervene there (Anonymous, 2025a;g;e;d; 2026b). MARINE (Zhao et al., 2025) injects object-level guidance from vision experts during generation. These are training-free and cheap, and they are complementary to what we do: they change the base model's error rate μ, while we control the error rate of what is emitted. §5 shows our layer composing on top of Leng et al. (2024).

Mitigation by training and by verification. A second family fine-tunes on hallucination-labelled data, using factually augmented RLHF (Sun et al., 2024a) or preference optimisation. A third generates then verifies: Woodpecker (Yin et al., 2024) corrects captions post hoc using detectors and a language model; REVERSE (Anonymous, 2025h) trains the model to emit confidence markers and then retrospectively resamples flagged spans; LURE (Zhou et al., 2024) rewrites captions identified as hallucination-prone. Knowledge-grounded variants retrieve from multimodal knowledge graphs (Lee et al., 2024) or medical corpora (Xia et al., 2024). REVERSE is the closest of these in spirit to our work, since it also *changes the output*; the difference is that it offers no risk guarantee, and its rewriting is learned rather than certified. Our own prior work (Dang et al., 2026) builds a verify-and-correct pipeline of this type, and the present paper can be read as asking what it would take to attach a guarantee to such a system.

2.2 Conformal prediction and risk control

Conformal prediction (Vovk et al., 2005) converts any score into prediction sets with finite-sample marginal coverage under exchangeability. Risk-controlling prediction sets (Bates et al., 2021) and Learn-Then-Test (Angelopoulos et al., 2021) generalise this to monotone risks and to selecting among many candidate configurations, the latter by multiple-testing arguments — Bonferroni, fixed-sequence testing, or graph-based procedures — so that a data-dependent choice of threshold remains valid. Bound choice matters in practice: Hoeffding's inequality is loosest, empirical Bernstein adapts to variance, and betting-style e-value constructions are tighter still (Waudby-Smith & Ramdas, 2024); for a binary loss the exact Clopper–Pearson interval (Clopper & Pearson, 1934) dominates all of them, which we verify in §7. Anytime-valid inference via e-processes and Ville's inequality (Waudby-Smith & Ramdas, 2024; Ramdas et al., 2023) extends validity to data-dependent stopping, which is the natural tool for sequence-level guarantees and which we discuss in §6.

Selective prediction itself is older (Chow, 1970; El-Yaniv & Wiener, 2010): the accept/abstain trade-off and its risk–coverage curve are classical, and conformal methods make the guarantee distribution-free.

2.3 Conformal methods for language and vision–language models

Conformal language modeling (Quach et al., 2024) calibrates stopping and rejection rules for sampled generations so that the returned set contains an acceptable output with high probability. Conformal factuality (Mohri & Hashimoto, 2024) decomposes an output into sub-claims and backs off until a factuality guarantee holds. ConfLVLM (Anonymous, 2025b) brings this to vision–language models: it treats each generated detail as a hypothesis, scores it with a heuristic uncertainty measure — CLIP similarity for scene descriptions, BiomedCLIP for radiology, LayoutLMv3 for documents — and filters, reporting a drop in claim error from 87.8% to 10.0% for LLaVA-1.5 scene descriptions. Token-level and next-token variants calibrate on entropy or logit statistics (Anonymous, 2025j;c). Online and bootstrapped variants address feedback and variance (Anonymous, 2025f;i). Pipeline-level joint coverage has also been studied (Anonymous, 2026d). Common to all of these is the operation performed on a flagged claim: *deletion, abstention, or back-off.*

Two very recent works are directly relevant. Anonymous (2026c) ask when conformal risk control can certify structured LLM outputs at all, prove the abstention floor we build on, establish a hierarchy of bounds (Hoeffding $\subseteq$ empirical Bernstein $\subseteq$ e-value CRC), and validate adaptive conformal inference under shift; they do not consider editing the output. Anonymous (2026a) propose budgeted conformal evidence acquisition: when a claim is borderline, acquire zoomed views, form a post-acquisition score, and re-threshold, with validity restored by calibrating on post-acquisition scores. Their setting (POPE/COCO, LLaVA and Qwen backbones), machinery (Clopper–Pearson with fixed-sequence testing) and motivation (avoid heavy abstention) coincide with ours, and they report coverage rising from 28% to 37% at $\alpha = 0.10$. The distinction is precise and important: BCEA *re-scores the model's original claim* and, in their words, makes no corrected answer. We substitute the answer. §5.5 shows the two mechanisms rescue different claims and compose.

2.4 Distribution shift caused by the predictor

Editing a generation changes the distribution of what follows. This is not exogenous covariate shift, for which weighted conformal prediction is the standard remedy (Tibshirani et al., 2019), but *feedback* covariate shift: the shift is a function of the deployed decision rule (Fannjiang et al., 2022). Validity can be restored in some such settings (Prinster et al., 2024), and adaptive conformal inference tracks drift online (Gibbs & Candès, 2021). Our sequential experiment (§6) shows that in generative correction this shift is not benign: it is accompanied by a measurable degradation in downstream faithfulness, which is a stronger obstacle than the loss of exchangeability alone.

3 Problem setup

3.1 Selective prediction with a risk guarantee

Let $(X, Y^\star)$ be a query–answer pair, where X contains an image and a question. A VLM produces $\hat{a}(X)$. Define the model's correctness indicator $Y = \mathbf{1}\{\hat{a}(X) = Y^\star\}$ and the *base risk* $\mu = \Pr[Y = 0]$. A selective predictor is a measurable rule that either emits an answer or abstains. Writing $\mathcal{E}$ for the (random) set of emitted items,

$$\mathcal{R}(\mathcal{E}) = \Pr[\text{emitted answer wrong} \mid \mathcal{E}], \qquad \mathcal{C}(\mathcal{E}) = \Pr[\text{item emitted}].$$

The goal is a policy with $\mathcal{R}(\mathcal{E}) \le \alpha$ certified at confidence $1 - \delta$, and $\mathcal{C}(\mathcal{E})$ as large as possible. We assume throughout:

Assumption 1 (Exchangeability)**.** Calibration and test items are exchangeable draws from the same distribution.

Assumption 2 (Pre-specified policy family)**.** The candidate policies form a family $\{\pi_\lambda\}_{\lambda \in \Lambda}$ whose index set and ordering are fixed before seeing calibration data.

3.2 Filtering, and the floor it cannot cross

A *filtering* policy uses a score $s : \mathcal{X} \to [0, 1]$, intended to increase with correctness, and emits $\hat{a}(X)$ when $s(X) \ge \tau$.

Proposition 1 (Abstention floor; Anonymous 2026c, Prop. 3)**.** *Let $\mu > \alpha$. Any selective predictor whose emitted answer is always $\hat{a}(X)$ satisfies*

$$1 - \mathcal{C}(\mathcal{E}) \;\ge\; \frac{\mu - \alpha}{1 - \alpha}$$

whenever $\mathcal{R}(\mathcal{E}) \le \alpha$. The bound is independent of the score s, of the concentration inequality used, and of the calibration sample size.

Two features of Proposition 1 drive this paper. It is *computable in advance*, since μ and α are known before any calibration; and it is *conditional on a premise* — that the emitted answer is the model's own — which is an assumption about the action space, not about the data.

3.3 Two channels

Definition 1 (Channels)**.** *Channel 1* is a correctness score $s(X) = \widehat{\Pr}[Y = 1 \mid \phi(X)]$ fitted on features $\phi(X)$ derived from the model's own output distribution together with grounding features. *Channel 2* is a separate predictor $b(X)$ that answers the query independently of the VLM, together with an evidence margin $m(X) \in [0, 1]$ measuring how decisively it does so. We write $Y^{(2)} = \mathbf{1}\{b(X) = Y^\star\}$ for channel 2's correctness.

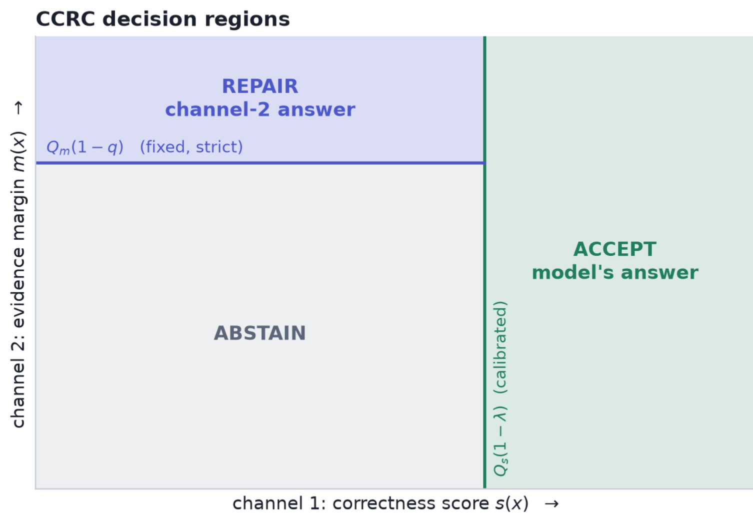

Figure 1: CCRC partitions the (s, m) plane. The acceptance threshold $Q_s(1-\lambda)$ is calibrated; the repair gate $Q_m(1-q)$ is fixed in advance and strict, so that as λ loosens the accepted region grows and the repair region shrinks. Keeping q fixed is what makes the family nested and lets a single fixed-sequence test suffice (§4.5).

In our instantiation channel 1 combines the VLM's yes/no probability, its confidence margin, and grounding features; channel 2 is an open-vocabulary object detector (Minderer et al., 2023) queried with the object named in the question, with m the absolute margin of its detection score around a fixed operating point. The essential property is that $Y^{(2)}$ is *not* a deterministic function of Y — the two channels can be right and wrong independently. §4.4 shows this is what makes repair possible.

4 Certified-correction risk control

4.1 The procedure

Fix a strict evidence quantile q (we use $q = 0.10$: repairs are drawn only from the top decile of channel-2 evidence). For permissiveness $\lambda \in [0, 1]$, policy π_λ acts as

$$\pi_\lambda(x) = \begin{cases} \textsc{accept } \hat{a}(x), & s(x) \geq Q_s(1-\lambda), \\ \textsc{repair with } b(x), & s(x) < Q_s(1-\lambda) \text{ and } m(x) \geq Q_m(1-q), \\ \textsc{abstain}, & \text{otherwise}, \end{cases} \tag{1}$$

where Q_s, Q_m are empirical quantiles computed on the calibration fold. Figure 1 shows the induced partition of the (s, m) plane. The emitted set is $\mathcal{E}_\lambda = A_\lambda \cup R_\lambda$ with A_λ the accepted and R_λ the repaired items, and the emitted-set error counts what is *actually emitted*,

$$E_\lambda = \sum_{i \in A_\lambda} (1 - Y_i) + \sum_{i \in R_\lambda} \left(1 - Y_i^{(2)}\right), \qquad K_\lambda = |A_\lambda| + |R_\lambda|. \tag{2}$$

4.2 Worked examples on real data

Before turning to aggregates, Figure 2 shows what the three decisions look like on individual POPE images, with the actual channel-1 score, channel-2 detection confidence, and resulting action for each. The three columns illustrate the regimes the procedure is designed to separate.

Figure 2: Real POPE images with the actual quantities CCRC uses. Each panel reports the question, the ground truth, LLaVA's answer with its probability, the detector's answer with its confidence and evidence margin, the channel-1 correctness score, and the resulting decision. **Accept** (left): both channels agree and the model is right. **Repair** (centre): the model misses an object that is genuinely present and receives a low correctness score, while the detector localises it with high confidence, so the answer is substituted and the error is fixed — items a filtering policy would abstain on. **Abstain** (right): the model is wrong and the detector is also wrong with weak evidence, so no certifiable answer exists and the procedure correctly declines. Thresholds are the calibrated $Q_s(1-\lambda)$ and the fixed gate $Q_m(1-q)$ from one split.

In the ACCEPT column the model answers a straightforward existence question confidently and correctly, the detector agrees, and the item is emitted unchanged; these cases carry the bulk of the coverage. The REPAIR column contains the cases that motivate the method: LLaVA is asked whether a computer mouse is present, answers "no" with $p = 0.21$, and is *wrong* — the mouse is there. Channel 1 correctly assigns a low correctness score ($s = 0.18$), which under a filtering policy would produce an abstention and no answer at all. The detector, however, localises the object with confidence 0.90, comfortably inside the top evidence decile, so CCRC substitutes the detector's answer and the item is emitted *correctly*. This is the dilution effect of §4.6 in a single instance: a high-precision repair converts an item that filtering would discard into a low-risk emitted item.

The ABSTAIN column shows the necessary complement. Here the model is wrong *and* the detector is also wrong, with weak evidence in both cases ($m = 0.04$ and $m = 0.03$); the questions ask about a handbag and a truck that are not present, and neither channel supplies a certifiable answer. Abstention is the correct action, and no amount of substitution would help. The existence of a substantial population of such items is why CCRC's gains are bounded by Proposition 3 rather than being able to recover the entire abstained mass.

4.3 Validity

Theorem 1 (CCRC validity). *Let Assumptions 1–2 hold, let $\{\lambda_1 < \cdots < \lambda_M\}$ be a pre-specified grid, and let*

$$U(e, k; \delta) = \sup\{p : \Pr[\mathrm{Bin}(k, p) \le e] > \delta\}$$

be the one-sided Clopper–Pearson upper confidence bound. Define $\hat{\lambda}$ by fixed-sequence testing: proceed through $j = 1, 2, \ldots$ while $U(E_{\lambda_j}, K_{\lambda_j}; \delta) \le \alpha$ on the calibration fold, and set $\hat{\lambda}$ to the last index that passed. Then the deployed policy $\pi_{\hat{\lambda}}$ satisfies

$$\Pr\big[\mathcal{R}(\mathcal{E}_{\hat{\lambda}}) \le \alpha\big] \;\ge\; 1 - \delta.$$

Proof. Fix j and consider the null $H_j : \mathcal{R}(\mathcal{E}_{\lambda_j}) > \alpha$. Conditional on $K_{\lambda_j} = k$, the emitted-set errors in (2) are, under Assumption 1, exchangeable Bernoulli indicators with mean $\mathcal{R}(\mathcal{E}_{\lambda_j})$; note that this holds for the mixture because each emitted item contributes exactly one indicator, namely $1 - Y_i$ if accepted and $1 - Y_i^{(2)}$ if repaired, and membership in A_{λ_j} or R_{λ_j} is determined by (s, m) and the pre-specified λ_j, q rather than by the labels. Hence $E_{\lambda_j} \sim \mathrm{Bin}(k, \mathcal{R}(\mathcal{E}_{\lambda_j}))$, and by construction of the Clopper–Pearson bound, $\Pr[U(E_{\lambda_j}, k; \delta) \le \alpha \mid H_j] \le \delta$: the event that the bound certifies a policy whose true risk exceeds α has probability at most δ, so $p_j = \mathbf{1}\{U \le \alpha\}$ is a valid level-δ test of H_j.

Because q is fixed, the family is nested in λ in the sense that A_λ is increasing and R_λ decreasing in λ, and the index set is pre-specified by Assumption 2. Fixed-sequence testing therefore applies: testing $H_1, H_2, \ldots$ in the pre-specified order and stopping at the first non-rejection controls the family-wise error rate at δ without any multiplicity correction (Angelopoulos et al., 2021). The returned $\hat{\lambda}$ is among the rejected hypotheses, so with probability at least $1 - \delta$ every rejected H_j is false, i.e. $\mathcal{R}(\mathcal{E}_{\hat{\lambda}}) \le \alpha$. ☐

Remark 1 (why the mixture is not a complication). The proof needs only that each emitted item contributes one $\{0, 1\}$ error indicator whose distribution does not depend on the labels of other items. Whether that indicator refers to the model's answer or to channel 2's is irrelevant. This is precisely why Proposition 1 does not bind: its premise restricts the *action space*, which the guarantee itself does not require.

Lemma 1 (Minimum certifiable emitted-set size). *With zero observed errors, $U(0, k; \delta) = 1 - \delta^{1/k}$. Hence no emitted set of size k can be certified at level α unless*

$$k \;\ge\; k_{\min} = \frac{\ln \delta}{\ln(1 - \alpha)}.$$

For $\alpha = \delta = 0.10$, $k_{\min} = \lceil 21.85 \rceil = 22$.

Proof. $\Pr[\mathrm{Bin}(k, p) \le 0] = (1 - p)^k$, so the upper bound with $e = 0$ solves $(1 - p)^k = \delta$, giving $U(0, k; \delta) = 1 - \delta^{1/k}$. This is decreasing in k, and $1 - \delta^{1/k} \le \alpha \iff \delta^{1/k} \ge 1 - \alpha \iff k^{-1} \ln \delta \ge \ln(1 - \alpha)$; dividing by the negative quantity $\ln(1 - \alpha)$ reverses the inequality and yields $k \ge \ln \delta / \ln(1 - \alpha)$. Since $U(e, k; \delta) \ge U(0, k; \delta)$ for $e \ge 0$, no error count can do better. ☐

Lemma 1 is elementary but consequential: a fixed sequence whose first policy emits fewer than $k_{\min}$ items fails at step one and, because fixed-sequence testing stops at the first non-rejection, returns *zero* coverage. We encountered exactly this failure and now begin the grid at the analytically derived minimum.

4.4 Repairs cannot be self-certified

The most economical design would avoid a second channel: if s says the model is probably wrong, emit the opposite. For binary answers this is not merely unreliable, it is circular.

Theorem 2 (Impossibility of self-certified repair). *Suppose the answer space is binary and the repair is the negation of the model's answer, so that $Y^{(2)} = 1 - Y$. Let $R \subseteq \mathcal{X}$ be any repair region. Then*

$$\mathcal{R}(R) = \Pr[Y = 1 \mid X \in R],$$

Table 1: Channel-2 (detector) accuracy by evidence-margin decile on POPE. Repairs are drawn only from the top decile, where accuracy is far above $1 - \alpha$. Accuracy degrades monotonically, which is why the repair gate must be strict.

Margin decile	10th (top)	9th	8th	6th	1st (bottom)
Detector accuracy	1.000	0.960	0.927	0.934	0.513

Table 2: Sensitivity to the repair gate q: change in certified coverage relative to filtering, over four settings $\times$ two risk levels. q is fixed a priori, not tuned.

Repair gate q	Outcome across the eight cells	Worst cell
0.10 (top decile)	positive in all eight (+0.5 to +8.0)	+0.5
0.25	mixed	−16.3
0.50	mixed	−15.5

so the repair region is certifiable at level α if and only if the model's accuracy *on R is at most α. Consequently, self-certified repair requires the score to identify a region of accuracy $\leq \alpha$; identifying a region of low* confidence *is not sufficient, and no monotone transformation of s can supply the missing information.*

Proof. On R the emitted answer is $1 - \hat{a}$, which is correct exactly when $\hat{a}$ is incorrect. Its error indicator is therefore $1 - Y^{(2)} = Y$, and averaging over R gives $\mathcal{R}(R) = \Pr[Y = 1 \mid X \in R]$. Certifiability at α requires $\mathcal{R}(R) \leq \alpha$, i.e. $\Pr[Y = 1 \mid X \in R] \leq \alpha$. The final claim follows because any repair rule of the form $\{s(X) \leq t\}$ has risk $\Pr[Y = 1 \mid s(X) \leq t]$, a functional of the joint law of (s, Y) that is invariant to monotone reparameterisation of s. □

Theorem 2 converts an intuition — "a hallucination score reports suspicion, not knowledge of the truth" — into a requirement. Empirically the requirement bites hard. On POPE with LLaVA-1.5, even in the bottom 2% of s the model's answer is still correct 13.3% of the time, exceeding $\alpha = 0.10$; in the bottom decile it is correct 41.3% of the time. No self-certified repair region exists at conventional risk levels, and a first implementation built this way repaired $\approx 0\%$ of items and underperformed plain filtering.

Channel 2 escapes this because $Y^{(2)}$ is a genuinely different random variable. Our detector is 93–100% accurate in its top evidence deciles (Table 1), so repairs drawn there are certifiable.

4.5 Why the repair gate must be decoupled

An earlier version of the procedure tied the repair gate to λ, i.e. used $m(x) \geq Q_m(1 - \lambda)$ in (1). This is natural but harmful: as λ loosens to admit more accepted items, the repair gate loosens in step and begins admitting low-precision repairs, so the fixed sequence terminates earlier. Empirically the coupled version gained on LLaVA (+4.7, +5.3 points) but *lost* on Qwen2-VL (−4.2) and on a VCD-decoded model (−2.9). It also required certifying two families (with and without repair) at $\delta/2$ and selecting between them, wasting half the confidence budget.

Fixing q removes both problems. The family becomes nested in the single parameter λ, so one fixed-sequence test at full δ suffices (Theorem 1), and repair precision is held constant while λ sweeps. Table 2 reports sensitivity: at $q = 0.10$ the procedure gains in all eight POPE-family cells, whereas looser gates can lose badly.

4.6 Why it gains: dilution and leverage

Our initial hypothesis was that repairs *spend* the accepted set's unused risk budget: filtering typically realises a risk below α (we measure 8.0% at $\alpha = 0.10$), and repairs with error above α could be admitted while the blend stayed compliant. The measured effect runs the other way, and is stronger.

Proposition 2 (Dilution relaxes the binding constraint)**.** *Fix λ and let the accepted set have mass c_a and risk r_a, and the repair set mass c_r and risk r_r. The emitted risk is the mixture $\bar{r} = (c_a r_a + c_r r_r)/(c_a + c_r)$. If $r_r < r_a$ then $\bar{r} < r_a$, strictly decreasing in c_r; consequently the feasible set $\{\lambda : \bar{r}(\lambda) \leq \alpha\}$ is a superset of the corresponding filtering feasible set, and the certified $\hat{\lambda}$ is at least as permissive. If instead $r_r > \alpha > r_a$, admitting repairs consumes slack and may tighten $\hat{\lambda}$.*

Proof. $\bar{r} - r_a = c_r(r_r - r_a)/(c_a + c_r)$, which is negative iff $r_r < r_a$, and $\partial\bar{r}/\partial c_r = c_a(r_r - r_a)/(c_a + c_r)^2$ has the same sign. Since the filtering policy is the special case $c_r = 0$, any λ feasible for filtering remains feasible when repairs with $r_r < r_a$ are added, and additional λ may become feasible. The last statement is the same computation with the inequality reversed. □

The empirically relevant regime is the first. Because repairs are drawn from the top evidence decile, their risk is *lower* than that of the acceptance gate's *marginal* items — the items that determine whether the constraint binds. Repairs therefore act as ballast: a small high-precision mass pulls $\bar{r}$ down and lets λ open further. The result is leverage. Figure 3B plots it: 0.9–1.8% of items repaired buys 1.9–4.7 points of certified coverage, a ratio of three to six.

4.7 When it helps: a checkable precondition

CCRC is not a free improvement. Its upside is bounded by what filtering leaves on the table, while its downside — perturbing a nearly-saturated constraint — is not symmetrically bounded.

Proposition 3 (Gain is capped by abstained mass)**.** *Let $\mathcal{C}_{\text{filt}}$ be the coverage certified by filtering at level (α, δ) and $\mathcal{C}_{\text{CCRC}}$ that certified by CCRC. Then*

$$\mathcal{C}_{\text{CCRC}} - \mathcal{C}_{\text{filt}} \;\leq\; 1 - \mathcal{C}_{\text{filt}} \;\leq\; \frac{\mu - \alpha}{1 - \alpha} + \varepsilon,$$

where $\varepsilon \geq 0$ is filtering's slack relative to the floor of Proposition 1. In particular, if $\mu \downarrow \alpha$ the achievable gain vanishes, while the loss from admitting repaired mass into a saturated constraint need not.

Proof. Coverage is at most one, so the gain cannot exceed the abstained fraction $1 - \mathcal{C}_{\text{filt}}$. By Proposition 1 applied to the filtering policy, $1 - \mathcal{C}_{\text{filt}} \geq (\mu - \alpha)/(1 - \alpha)$; writing the excess as ε gives the stated inequality. As $\mu \to \alpha^+$ the floor tends to zero, so $1 - \mathcal{C}_{\text{filt}} \to \varepsilon$, which is small whenever filtering is near-optimal. The asymmetry claim follows because Proposition 2 shows the certified $\hat{\lambda}$ can strictly decrease when repairs are admitted with $r_r > r_a$, and no corresponding lower bound on $\mathcal{C}_{\text{CCRC}}$ holds. □

Proposition 3 predicts our one negative result. On AMBER's discriminative split, $\mu = 11.4\%$ against $\alpha = 10\%$: the floor is 1.6%, filtering already certifies 92.1%, and the theoretical headroom is at most 7.9 points. Measured, CCRC *loses* 7.5 points there. Figure 3A shows the gain tracking $\mu - \alpha$ across all settings. Both quantities are known before calibration, so the decision to deploy CCRC can be made a priori.

4.8 Algorithm

5 Experiments

Protocol. Every reported number uses a three-way disjoint split — fit the score, calibrate the threshold, test — with fixed-sequence testing and Clopper–Pearson bounds at $\delta = 0.10$, averaged over 20–100 random splits. We report realised test risk beside coverage in every table so validity can be audited. The three-way split is not cosmetic: §7 shows that reusing the calibration fold to fit the score silently breaks the guarantee for high-capacity scores.

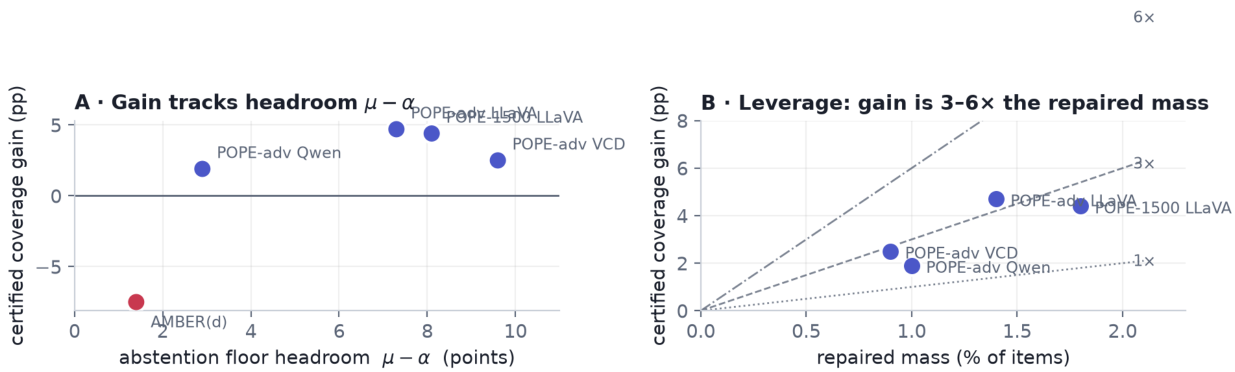

Figure 3: (A) Certified coverage gain against the abstention-floor headroom $\mu - \alpha$. Gains appear when the headroom exceeds roughly three points and reverse when it does not: AMBER(d), with 1.4 points of headroom, is the single loss. (B) Leverage: gain plotted against repaired mass, with 1×, 3× and 6× reference lines. A small high-precision repaired mass buys several times its own size in coverage, as Proposition 2 predicts.

Algorithm 1 CCRC (calibration and deployment)

Require: folds $\mathcal{D}_{\text{fit}}, \mathcal{D}_{\text{cal}}$ (disjoint), levels α, δ, repair gate q, grid $\lambda_1 < \cdots < \lambda_M$ with λ_1 chosen so $K_{\lambda_1} \geq k_{\min}$ (Lemma 1)

fit channel-1 score s on $\mathcal{D}_{\text{fit}}$ ▷ disjoint from calibration; see §7
$t_m \leftarrow Q_m(1-q)$ on $\mathcal{D}_{\text{cal}}$ ▷ repair gate fixed
$\hat{\lambda} \leftarrow \varnothing$
for $j = 1$ to M **do**
  $A \leftarrow \{i \in \mathcal{D}_{\text{cal}} : s_i \geq Q_s(1-\lambda_j)\}$; $R \leftarrow \{i \notin A : m_i \geq t_m\}$
  $E \leftarrow \sum_{i \in A}(1-Y_i) + \sum_{i \in R}(1-Y_i^{(2)})$; $K \leftarrow |A| + |R|$
  if $K > 0$ and $U(E, K; \delta) \leq \alpha$ **then** $\hat{\lambda} \leftarrow \lambda_j$
  else break ▷ fixed-sequence testing
  end if
end for
at test time, for input x**: if** $s(x) \geq Q_s(1-\hat{\lambda})$ emit $\hat{a}(x)$; **else if** $m(x) \geq t_m$ emit $b(x)$; **else** abstain

Models and data. Backbones are LLaVA-1.5-7B (Liu et al., 2024) in 4-bit and Qwen2-VL-2B-Instruct (Wang et al., 2024); we additionally evaluate LLaVA decoded with visual contrastive decoding (Leng et al., 2024). Channel 2 is OWLv2-base (Minderer et al., 2023). Benchmarks are POPE (Li et al., 2023) (random and adversarial splits), AMBER discriminative (Wang et al., 2023), MME (Fu et al., 2023), GQA (Hudson & Manning, 2019) and HallusionBench (Guan et al., 2024). All per-item extracted signals are released so that the analyses can be reproduced without a GPU.

5.1 The nonconformity score

Validity holds for any score; only efficiency depends on which one is used. We therefore first characterise what makes a score efficient here.

Table 3 compares scores on POPE. A learned combination of the model's own confidence signals is a solid baseline. Generic image–text similarity (Radford et al., 2021) adds little. An open-vocabulary *detection* signal adds a great deal: +8.4 points of certified coverage against +2.3 for CLIP.

Against a baseline given *both* confidence and multi-sample self-consistency — the scoring philosophy of Anonymous (2025b) — detection grounding still adds 10.2 ± 8.6 points on the adversarial split (Table 4).

Table 3: Selective prediction on POPE (LLaVA-1.5-7B, $n = 1500$). AURC is area under the risk–coverage curve (lower better). Realised test risk was $\leq \alpha$ in every row.

Score	AURC ↓	cov@10% ↑	AUROC ↑
Raw confidence	0.0775	62.1%	0.767
Learned, confidence only	0.0681	65.2%	0.786
+ CLIP grounding (Radford et al., 2021)	0.0632	67.5%	0.800
+ detection grounding (Minderer et al., 2023)	**0.0546**	**73.6%**	**0.835**
+ both	0.0538	74.9%	0.838

Table 4: Against a faithful self-consistency baseline on POPE-adversarial, the hardest split ($K = 3$ samples, LLaVA-1.5, $n = 450$).

Score	AURC ↓	cov@10% ↑
Self-consistency only	0.135	0.0%
Confidence + self-consistency	0.077	61.8%
+ detection grounding (ours)	**0.059**	**72.0%**

Two regularities follow. The advantage of grounding *grows as the guarantee tightens*: at $\alpha = 0.05$ it nearly doubles usable coverage (30.0% $\to$ 52.5%), while at $\alpha = 0.20$ both scores certify almost everything (Figure 4). And the advantage tracks how much the base model hallucinates: on the more accurate Qwen2-VL-2B the same addition is worth 1.9 points rather than 10.2. Grounding is most valuable exactly where selective prediction is most expensive.

Remark 2 (accuracy is not separability). A detector whose standalone accuracy matches the VLM's on POPE (83.1% vs. 81.9%) certifies far less coverage when used as a selective predictor in its own right (55.3% vs. 68.2%). Conformal coverage depends on how well a score *separates* correct from incorrect outputs, not on how often the underlying predictor is right. This also answers the natural objection to CCRC — "if channel 2 is good, why not just use it?" — in three of four settings (Table 7).

5.2 CCRC across settings

Table 5 reports CCRC against filtering under an identical protocol. The guarantee holds in every row. Gains are +1.9 to +8.0 points where the precondition of §4.7 is satisfied, and the single violation of that precondition is also the single loss.

Isolating the cause of the AMBER loss. Two candidate explanations were tested and rejected. It is not a weak repair channel: the detector is 95.6% accurate on that dataset overall and 100% accurate in the top-decile region repairs are drawn from. It is not small sample size: subsampling POPE to the same $n = 228$ still gains +3.2 points, and the gain grows with n (+5.6 at 400, +10.3 at 700). The cause is the absence of headroom, exactly as Proposition 3 predicts.

5.3 Where grounding does not apply

Benchmark composition determines whether channel 2 exists at all. Table 6 reports the full sweep. On MME's full split, GQA's yes/no subset and HallusionBench, the questions are largely about attributes, counting or reasoning, so no object can be extracted for the detector to check; the procedure reduces to filtering, which remains valid. HallusionBench is instructive: LLaVA scores 51.2%, essentially chance, and the certified coverage is 0.0%. That is the guarantee working as intended — with $\mu \approx 0.49$ and $\alpha = 0.10$, Proposition 1 forces abstention on at least 43% of inputs, and in practice on all of them.

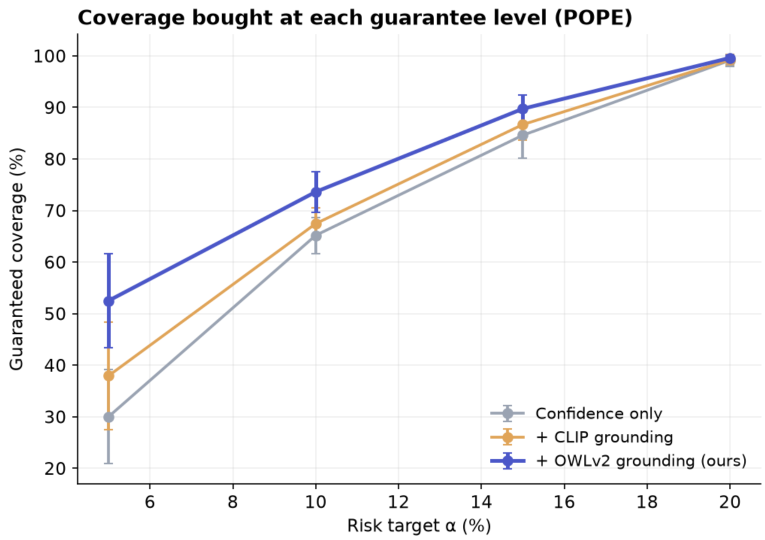

Figure 4: Certified coverage as a function of the risk target. The grounded score dominates at every level and the margin widens as α tightens; at $\alpha = 0.05$ it nearly doubles coverage. Error bars are one standard deviation over splits.

Table 5: CCRC (q=0.10) versus filtering. Realised test risk was $\leq \alpha$ in every row; the risk column reports CCRC's. The final row violates the precondition of Proposition 3 and is the only loss.

Dataset	Backbone	n	μ	α	Filter	CCRC	Gain
POPE	LLaVA-1.5	1500	18.1%	0.10	68.2%	**72.6%**	+4.4
POPE-adv	LLaVA-1.5	444	17.3%	0.10	42.1%	**46.8%**	+4.7
POPE-adv	Qwen2-VL-2B	591	12.9%	0.10	77.5%	**79.4%**	+1.9
POPE-adv	LLaVA+VCD	591	19.6%	0.10	45.0%	**47.5%**	+2.5
POPE	LLaVA-1.5	1500	18.1%	0.15	86.8%	**88.4%**	+1.7
POPE-adv	LLaVA-1.5	444	17.3%	0.15	69.2%	**77.2%**	+8.0
POPE-adv	Qwen2-VL-2B	591	12.9%	0.15	94.4%	**94.9%**	+0.5
POPE-adv	LLaVA+VCD	591	19.6%	0.15	73.4%	**75.4%**	+2.0
AMBER(d)	LLaVA-1.5	228	11.4%	0.10	92.1%	84.5%	−7.5

5.4 Composition with a mitigation decoder

Because the two programmes of §2 are orthogonal — mitigation lowers μ, risk control governs what is emitted — our layer should compose with a mitigation decoder. It does, and most strongly where the decoder's own confidence is poorly calibrated. On POPE-adversarial with visual contrastive decoding (Leng et al., 2024), adding grounding improves AURC from 0.1030 to 0.0603 and certified coverage from 47.6% to 64.2% ($\Delta\text{AURC} = 0.043 \pm 0.007$).

5.5 Baselines and attribution

Detector alone. Table 7 answers Remark 2's objection directly. CCRC beats a detector-only selective predictor in three of four settings, sometimes by very large margins, because the detector's correctness is harder to *score* than the VLM's. Where the detector genuinely dominates the VLM in accuracy (AMBER: 95.6% vs. 88.6%) it also wins standalone, which is a second reason CCRC is the wrong tool in that regime.

Table 6: Full dataset sweep. "grnd" indicates whether channel 2 applies. Coverage columns are confidence-only → with grounding, at $\alpha = 0.10$.

Dataset	Backbone	acc	grnd	cov@10%
POPE (1500)	LLaVA-1.5	81.9%	yes	63.6 → **73.1**%
POPE-adv	LLaVA-1.5	82.7%	yes	46.6 → **57.4**%
POPE-adv	Qwen2-VL-2B	87.3%	yes	83.5 → **85.5**%
POPE-adv	LLaVA+VCD	80.3%	yes	21.8 → **53.9**%
MME-existence[a]	LLaVA-1.5	95.0%	yes	underpowered
AMBER(d)	LLaVA-1.5	78.6%	partial	see Table 5
MME (full)	LLaVA-1.5	68.9%	no	5.7%
GQA (yes/no)	LLaVA-1.5	72.4%	no	17.8%
HallusionBench	LLaVA-1.5	51.2%	no	0.0%

[a] Only 60 items; below the sample size at which a 10% risk target can be certified for a set of this composition (cf. Lemma 1).

Table 7: Detector-only selective prediction versus CCRC ($\alpha = 0.10$, identical protocol).

Setting	VLM acc	det. acc	Filter (VLM)	CCRC	Detector only
POPE-1500 LLaVA	81.9%	83.1%	68.2%	**72.6%**	55.3%
POPE-adv LLaVA	82.7%	82.4%	42.1%	**46.8%**	13.0%
POPE-adv Qwen2-VL	87.1%	82.9%	77.5%	**79.4%**	21.8%
AMBER(d) LLaVA	88.6%	95.6%	92.1%	84.5%	**92.6%**

A published scorer, and where our margin comes from. Table 8 places the CLIP-similarity scoring function of Anonymous (2025b) in our pipeline and adds our components one at a time. The aggregate margin is +67 points, and reporting only that number would misattribute it: +62.6 of the 67 comes from the score, and +4.4 from certified correction. We also note the comparison understates Anonymous (2025b), whose method was designed for free-form caption claims rather than yes/no probes; we transplant their scorer, not their system. Figure 5 shows the corresponding risk–coverage curves.

The closest concurrent method. Anonymous (2026a) rescue borderline claims by acquiring zoomed views and re-scoring the *original* assertion. Holding the base score fixed and adding one mechanism per arm (Table 9), the two mechanisms are complementary: composing them beats every single-mechanism arm at both risk levels. Re-reading the image and replacing the answer rescue different claims. We do not claim to beat Anonymous (2026a): our reproduction uses three coarse crops and under-delivers relative to their published gain at $\alpha = 0.10$, so the defensible conclusion is composition, not superiority.

6 A sequential failure mode

Correction is more attractive in free-form generation than in yes/no answering: replacing a hallucinated word preserves a caption that deletion would mutilate. The extension looks mechanical — decompose the caption into claims, repair the unsupported ones, continue decoding — and two obstacles are already known. Editing a prefix makes later tokens off-policy, which is feedback covariate shift rather than exogenous shift (Fannjiang et al., 2022), so weighted conformal methods (Tibshirani et al., 2019) do not apply. And per-claim guarantees say nothing about a whole sequence, whose length is itself data-dependent.

Both could in principle be handled. Prefix-conditional (Mondrian) p-values make each decision valid given the realised prefix, so how we arrived there — including by our own edits — is absorbed into the conditioning; e-processes with Ville's inequality give sequence-level anytime validity, and are robust to adaptive intervention because our corrections are predictable with respect to the decoding filtration (Waudby-Smith & Ramdas,

Table 8: Attribution on POPE ($\alpha = 0.10$). Most of the margin over a published scorer comes from the score, not from our procedure.

Method	Coverage	Δ
CLIP-similarity scorer (Anonymous, 2025b), filtering	5.6%	—
+ VLM confidence	41.5%	+35.9
+ detection grounding	68.2%	+26.7
+ certified correction (CCRC)	**72.6%**	+4.4

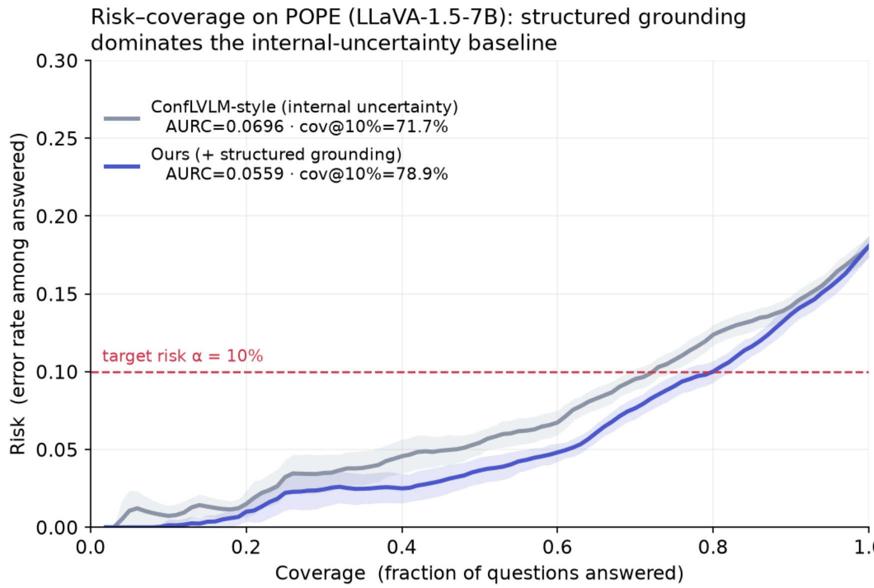

Figure 5: Risk–coverage curves on POPE. The grounded score (blue) lies below the internal-uncertainty baseline at every coverage level, i.e. lower error for the same coverage and more coverage for the same error target. Shaded bands are one standard deviation over 20 splits; the dashed line marks $\alpha = 0.10$.

2024; Ramdas et al., 2023). The residual difficulty is that the calibration pool drifts, since the deployed policy writes different text. The natural fix is on-policy calibration by iteration to a fixed point, and its convergence argument requires the correction map to be monotone. We measured whether it is, before attempting a proof.

Design. For each image we generate a caption greedily, locate the first hallucinated object mention at position t using AMBER's per-image present/absent object lists, and construct two prefixes *identical up to* t: one retaining the hallucinated word, one substituting the best-detected object the image genuinely contains. We then continue decoding the same number of tokens from each prefix and count hallucinated mentions in the continuations only. Matched prefixes make the contrast causal: the only difference between the two continuations is the intervention.

Result. Repair makes the continuation worse (Table 10, Figure 6). Downstream hallucinated mentions rise from 1.207 to 1.413, a paired difference of $+0.207 \pm 0.091$ ($n = 121$, $t = 2.27$, $p = 0.025$; Wilcoxon $p = 0.035$), with 45 cases worsening against 25 improving.

What this rules out. The correction map is not monotone, so the fixed-point argument has no basis: monotone convergence arguments of Knaster–Tarski type do not apply, and iterating calibration and deployment need not converge. We therefore do not claim a sequential guarantee of that form. What survives is a coupling bound, which is weaker but assumption-light.

Table 9: One common base score, one mechanism added per arm (POPE, LLaVA-1.5, $n = 394$ grounded items). All arms valid. Composition beats every single mechanism.

Arm	cov@0.10	cov@0.15
Base filter (confidence only)	5.5%	16.5%
+ evidence acquisition (Anonymous, 2026a)	5.5%	24.1%
+ detection-grounded score (ours)	32.1%	63.2%
+ certified correction (ours)	34.5%	71.8%
Composed	**35.8%**	**74.2%**

Table 10: Paired matched-prefix test of monotonicity ($n = 121$ cases). Repair significantly increases hallucination in the continuation.

Downstream measure (continuation only)	Original prefix	Repaired prefix
Hallucinated mentions (mean)	1.207	**1.413**
Hallucinations per 100 words	3.61	4.30
Faithful mentions (mean)	1.579	1.752
Continuation length (words)	33.4	32.9
Paired difference $+0.207 \pm 0.091$; $t = 2.27$, $p = 0.025$; Wilcoxon $p = 0.035$		
Cases better / worse / tie: 25 / 45 / 51		

Theorem 3 (Coupling bound for corrected outputs)**.** *Let O denote the original output and O' the output after a repair rule is applied to a region R. Then*

$$\mathcal{R}(O') \;\le\; \mathcal{R}(O) \;+\; \Pr[X \in R] \cdot \Pr[\textit{repair wrong} \mid X \in R].$$

Consequently, controlling $\mathcal{R}(O) \le \alpha_1$ and the second term by α_2 with confidence $1-\delta/2$ each yields $\mathcal{R}(O') \le \alpha_1 + \alpha_2$ with confidence $1 - \delta$.

Proof. Decompose by whether the item is repaired. On R^c the output is unchanged, contributing at most $\mathcal{R}(O)$. On R the emitted answer is wrong only if the repair is wrong, so the contribution is at most $\Pr[X \in R]\Pr[\text{repair wrong} \mid X \in R]$. Summing gives the bound; the two-level statement follows by a union bound over the two events on which the respective certifications fail, each of probability at most $\delta/2$. □

What this implies for algorithm design. Beyond closing off a proof strategy, the measurement identifies a cost that a myopic procedure does not charge for. Greedy per-claim repair optimises the risk of the claim in front of it while imposing roughly 0.21 additional hallucinated claims on the continuation. A correct sequential procedure must therefore price the *continuation*, not the claim — for instance by searching over rewrite trajectories under an accumulated risk ledger rather than deciding each claim independently. We regard this as the main open problem the paper leaves, and note that it is a genuinely different statistical object: because repairing claim i changes which claims $i+1, \dots$ exist, the item set is endogenous, and selecting among trajectories introduces a selective-inference problem that per-item conformal methods do not face.

Scope of the negative result. Our repair substitutes the globally best-detected object, which can be contextually inappropriate mid-sentence and may push the prefix off-policy for reasons unrelated to faithfulness. The supported claim is that *context-blind* substitution carries a downstream cost, not that all correction must. Constraining repairs to the model's own top-k continuations — which the one-shot procedure of §4 already implies — is the natural remedy and remains to be tested.

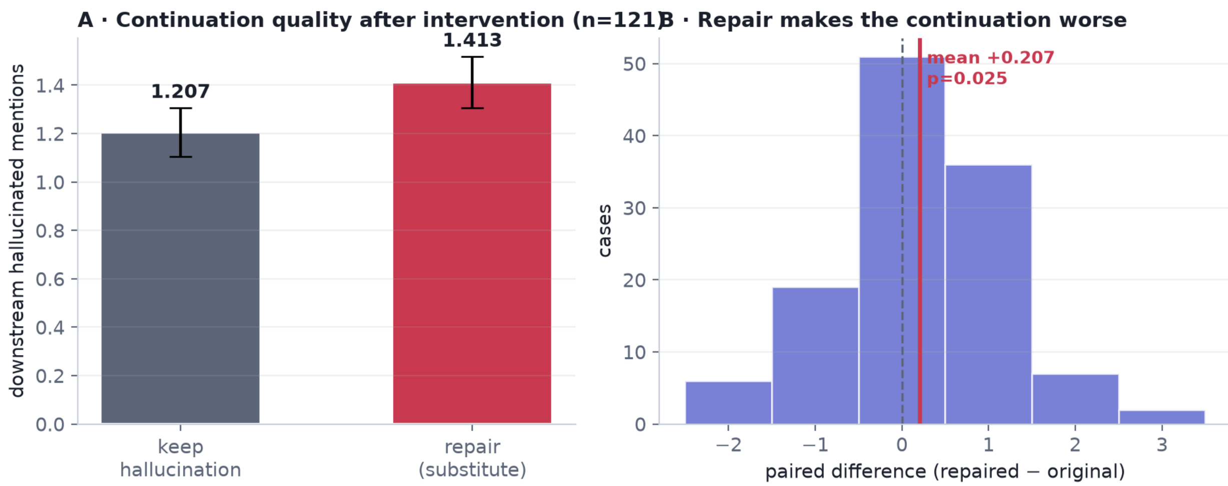

Figure 6: (A) Mean hallucinated mentions in the continuation after keeping versus repairing the flagged claim; error bars are standard errors. (B) Distribution of the paired difference; mass to the right of zero indicates repair worsening the continuation. The effect is modest in size but statistically clear, and its sign is what matters for the theory.

Table 11: Combiner ablation on POPE ($\alpha = 0.10$, 30 repetitions). Under data reuse, high-capacity combiners *break* the guarantee; under a three-way split they are valid and tied. Logistic regression is chosen deliberately.

	Naive: fit = calibrate		Three-way split	
Combiner	cov@10%	test risk	cov@10%	test risk
Logistic regression	74.8%	8.8%	72.5%	8.4%
Gradient boosting	91.3%	**13.3%** (violated)	72.5%	8.3%
Random forest	88.5%	**12.1%** (violated)	75.7%	8.5%
MLP $(64, 32)$	≈28%	—	≈28%	—

7 Ablations

7.1 The score combiner, and a validity trap

Channel 1 is a learned function of a handful of scalars. It is natural to ask whether a more expressive combiner helps. It does not, and it can silently invalidate the guarantee.

Table 11 reports two protocols. Under the naive protocol, where the combiner is fitted on the same fold used to calibrate τ, gradient boosting and random forests appear to certify far more coverage — 91.3% and 88.5% — but their *realised* test risk is 13.3% and 12.1% against a 10% target. The mechanism is straightforward: a high-capacity model overfits the calibration scores, so the empirical error on the calibration fold understates the true error, and the threshold is set too permissively. Under a three-way split all combiners are statistically tied and all are valid, while a small multilayer perceptron collapses ($\approx$ 28% coverage, high variance) from overfitting four features.

We draw two conclusions. Practically, the score must be fitted on a fold disjoint from calibration — a discipline that costs coverage but is necessary for any scorer whose capacity is not negligible. Conceptually, capacity is not the bottleneck: the features are already informative scalars, and what limits efficiency is the quality of the grounding signal, not the flexibility of the combiner.

Table 12: Risk-bound ablation (POPE, $\alpha = 0.10$, three-way split). Exact binomial dominates for a binary loss; the others are valid but conservative.

Upper confidence bound	cov@10%	realised risk
Clopper–Pearson (exact)	**72.5%**	8.4%
Empirical Bernstein	54.9%	4.6%
Hoeffding	38.3%	3.2%

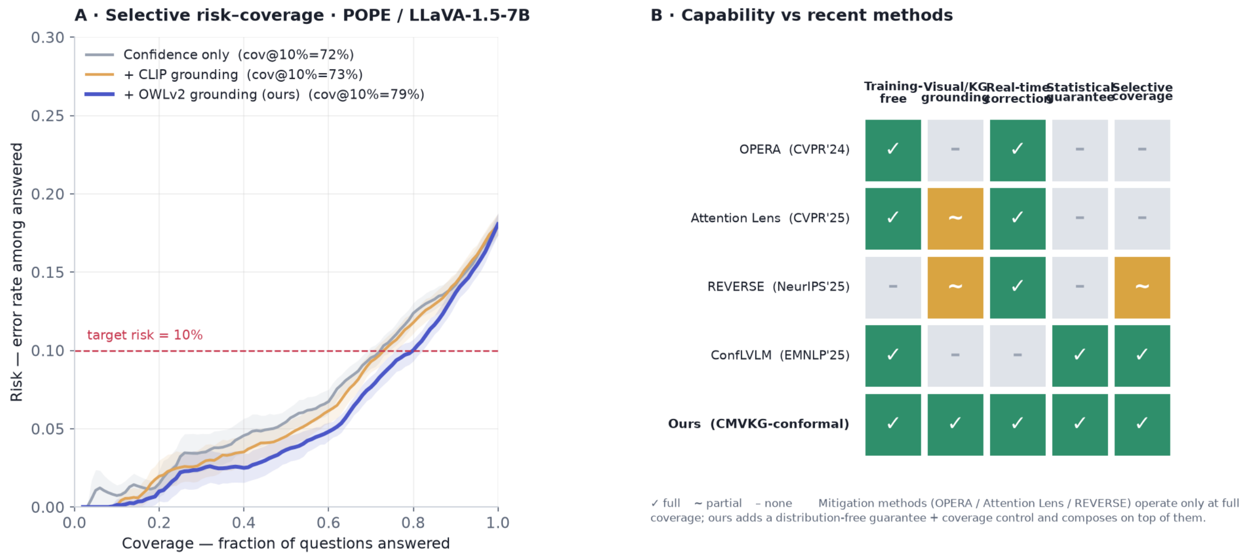

Figure 7: (A) Risk–coverage on POPE for the score variants. (B) Capability comparison with recent methods. Mitigation approaches change the base model but provide no guarantee; existing conformal approaches guarantee but only remove or abstain. Our layer composes on top of mitigation decoders (§5).

7.2 The risk bound

The emitted-set error is a binomial proportion, so an exact interval should dominate general-purpose concentration inequalities, and it does: at $\alpha = 0.10$ Clopper–Pearson certifies 72.5% coverage, empirical Bernstein 54.9%, and Hoeffding 38.3% (Table 12). All three are valid; the latter two are simply conservative, realising 4.6% and 3.2% risk against a 10% budget. This is consistent with the hierarchy established by Anonymous (2026c) and indicates that for binary losses the choice of bound is worth more than several points of coverage. Variance-adaptive or betting-style bounds (Waudby-Smith & Ramdas, 2024) become preferable for graded losses, which is the natural setting for claim-level factuality scores.

7.3 Positioning summary

Figure 7 summarises where CCRC sits. Panel A gives risk–coverage curves; panel B contrasts capabilities across recent methods. Mitigation methods (Huang et al., 2024; Anonymous, 2025a;h) operate only at full coverage and offer no guarantee; conformal methods (Anonymous, 2025b;f) guarantee but only delete or abstain; Anonymous (2026a) re-scores without changing the answer. CCRC is, to our knowledge, the first to certify an output that differs from the model's — while requiring, as Theorem 2 shows, an independent channel to do so.

8 Discussion

What the results support. Permitting a selective predictor to emit a substituted answer is a real degree of freedom: it evades the premise of Proposition 1 and, when the preconditions hold, converts a small amount of high-precision external evidence into several times as much certified coverage. The guarantee is unaffected by the substitution (Theorem 1), because risk control cares about error indicators rather than about provenance.

What they do not. The gains are modest and conditional. The mechanism requires headroom (Proposition 3), a strict gate (Table 2), and an independent channel (Theorem 2); each is a real constraint, and one of our five benchmarks fails the first. Most of our margin over a published baseline comes from the nonconformity score rather than from the algorithm (Table 8), and the closest concurrent method is complementary rather than dominated (Table 9). The one-shot theory is a correct but standard application of fixed-sequence testing with exact bounds; the interesting theory is sequential, and §6 shows the most natural route to it is closed.

Limitations. Channel 2 is an object detector, so the evaluated method addresses existence hallucination; on attribute-, counting- and reasoning-heavy benchmarks it degenerates to filtering, which remains valid but gains nothing. The independence of channel 2 is an assumption, and a detector sharing the VLM's blind spots would weaken certification without invalidating it. The repair gate q is fixed a priori rather than learned, and a data-driven choice would require its own share of the confidence budget. Exchangeability is assumed throughout; under deployment shift the stated level need not hold, and adaptive conformal methods (Gibbs & Candès, 2021) would be required. Finally, our sequential experiment uses one backbone and a context-blind repair rule, and its negative conclusion should be read with that scope.

Outlook. The open problem we consider most interesting is the one the negative result exposes. Because repairing a claim changes which subsequent claims exist, a sequential procedure faces an endogenous item set and must select among rewrite trajectories rather than filter a fixed list. Selecting the best trajectory and then reporting its certificate is a selective-inference error, so a correct procedure needs a certificate that is simultaneous over the search tree. Building that, and pricing the downstream cost we measured, is the natural continuation of this work.

9 Conclusion

Selective prediction for vision–language models is expensive because of a bound whose premise — that the emitted answer is the model's own — is an assumption about the action space rather than about the data. Relaxing it works, but not freely: certified correction requires an independent evidence channel, a strict repair gate, and base risk comfortably above the target, and it delivers its gains through a dilution effect that gives small repairs disproportionate leverage. Applied myopically in generation, the same idea makes things measurably worse. We have tried to report the conditions and the failures as carefully as the improvements, in the belief that for methods whose selling point is a guarantee, the boundaries of that guarantee are the substance of the contribution.

Broader Impact Statement

Attaching guarantees to model outputs can improve safety, but a guarantee is only as meaningful as its assumptions. Our procedure controls the error rate among emitted answers under exchangeability and requires a genuinely independent evidence channel; if either fails — under deployment shift, or with a detector sharing the model's blind spots — the stated level may not hold. Emitting a *corrected* answer also changes the failure mode presented to a user, replacing an abstention that invites scrutiny with a confident substitution that may not. For this reason we regard the checkable precondition and the sequential negative result as safety-relevant components of the contribution rather than limitations to be minimised, and we recommend that any deployment verify the precondition and monitor channel independence rather than treating the nominal α as a guarantee that transfers unconditionally.

Acknowledgments

Placeholder for the camera-ready version.

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A Additional results

A.1 Synthetic validation of the three pillars

Before any real-data experiment we validated the procedure on synthetic data where ground truth is controllable, confirming (i) that the certified risk is respected, (ii) that a better-separated score buys coverage at fixed risk, and (iii) that a single global threshold can satisfy a marginal guarantee while violating it on a minority subgroup (25.3% realised risk against a 10% target), which group-conditional (Mondrian) calibration repairs (9.8% and 9.7% on the two groups). The last point motivates treating conditional coverage as a separate requirement from marginal coverage; we report it on synthetic data only, as our real-data samples are not category-balanced.

A.2 Multi-α table

Table 13 gives certified coverage at four risk levels, the numerical version of Figure 4.

Table 13: Certified coverage at four risk targets (POPE, LLaVA-1.5, 20 splits).

Score	$\alpha = 0.05$	0.10	0.15	0.20
Confidence only	30.0%	65.2%	84.6%	99.1%
+ detection grounding	**52.5%**	**73.6%**	**89.7%**	**99.6%**

A.3 Reproducibility

All extracted per-item signals (model probabilities, self-consistency counts, detector scores, labels) are released as JSON, so every table and figure in this paper can be regenerated on CPU without access to a GPU. The GPU scripts that produced those signals are released as well, together with the failed variants discussed in §4.5 and §4.4, which we retain because the failures are part of the argument.